FANUC Robot series

R-30*i*B AND R-30*i*B PLUS CONTROLLER

Servo Gun Tip Wear

USER'S GUIDE

Version 8.20 and later

MARUBSGTD09131E REV. B

FANUC America Corporation
3900 W. Hamlin Road
Rochester Hills, Michigan 48309–3253

Copyrights and Trademarks

The descriptions and specifications contained in this manual were in effect at the time this manual was approved for printing. FANUC America Corporation, hereinafter referred to as FANUC, reserves the right to discontinue models at any time or to change specifications or design without notice and without incurring obligations.

FANUC manuals present descriptions, specifications, drawings, schematics, bills of material, parts, connections and/or procedures for installing, disassembling, connecting, operating and programming FANUC products and/or systems. Such systems consist of robots, extended axes, robot controllers, application software, the KAREL® programming language, INSIGHT® vision equipment, and special tools.

FANUC recommends that only persons who have been trained in one or more approved FANUC Training Course(s) be permitted to install, operate, use, perform procedures on, repair, and/or maintain FANUC products and/or systems and their respective components. Approved training necessitates that the courses selected be relevant to the type of system installed and application performed at the customer site.

WARNING

This equipment generates, uses, and can radiate radiofrequency energy and if not installed and used in accordance with the instruction manual, may cause interference to radio communications. As temporarily permitted by regulation, it has not been tested for compliance with the limits for Class A computing devices pursuant to Subpart J and Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference. Operation of the equipment in a residential area is likely to cause interference, in which case the user, at his own expense, will be required to take whatever measure may be required to correct the interference.

FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

FANUC America Corporation
Training Department
3900 W. Hamlin Road
Rochester Hills, Michigan 48309-3253
www.fanucamerica.com

For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 1-800-47-ROBOT (1-800-477-6268). International customers should call 011-1-248-377-7159.

Send your comments and suggestions about this manual to:
product.documentation@fanucamerica.com

Patents

One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

4,630,567 4,639,878 4,707,647 4,708,175 4,708,580 4,942,539 4,984,745 5,238,029 5,239,739 5,272,805 5,293,107 5,293,911 5,331,264 5,367,944 5,373,221 5,421,218 5,434,489 5,644,898 5,670,202 5,696,687 5,737,218 5,823,389 5,853,027 5,887,800 5,941,679 5,959,425 5,987,726 6,059,092 6,064,168 6,070,109 6,086,294 6,122,062 6,147,323 6,204,620 6,243,621 6,253,799 6,285,920 6,313,595 6,325,302 6,345,818 6,356,807 6,360,143 6,378,190 6,385,508 6,425,177 6,477,913 6,490,369 6,518,980 6,540,104 6,541,757 6,560,513 6,569,258 6,612,449 6,703,079 6,705,361 6,726,773 6,768,078 6,845,295 6,945,483 7,149,606 7,149,606 7,211,978 7,266,422 7,399,363

FANUC CORPORATION Patent List

4,571,694 4,626,756 4,700,118 4,706,001 4,728,872 4,732,526 4,742,207 4,835,362 4,894,596 4,899,095 4,920,248 4,931,617 4,934,504 4,956,594 4,967,125 4,969,109 4,970,370 4,970,448 4,979,127 5,004,968 5,006,035 5,008,834 5,063,281 5,066,847 5,066,902 5,093,552 5,107,716 5,111,019 5,130,515 5,136,223 5,151,608 5,170,109 5,189,351 5,267,483 5,274,360 5,292,066 5,300,868 5,304,906 5,313,563 5,319,443 5,325,467 5,327,057 5,329,469 5,333,242 5,337,148 5,371,452 5,375,480 5,418,441 5,432,316 5,440,213 5,442,155 5,444,612 5,449,875 5,451,850 5,461,478 5,463,297 5,467,003 5,471,312 5,479,078 5,485,389 5,485,552 5,486,679 5,489,758 5,493,192 5,504,766 5,511,007 5,520,062 5,528,013 5,532,924 5,548,194 5,552,687 5,558,196 5,561,742 5,570,187 5,570,190 5,572,103 5,581,167 5,582,750 5,587,635 5,600,759 5,608,299 5,608,618 5,624,588 5,630,955 5,637,969 5,639,204 5,641,415 5,650,078 5,658,121 5,668,628 5,687,295 5,691,615 5,698,121 5,708,342 5,715,375 5,719,479 5,727,132 5,742,138 5,742,144 5,748,854 5,749,058 5,760,560 5,773,950 5,783,922 5,799,135 5,812,408 5,841,257 5,845,053 5,872,894 5,887,122 5,911,892 5,912,540 5,920,678 5,937,143 5,980,082 5,983,744 5,987,591 5,988,850 6,023,044 6,032,086 6,040,554 6,059,169 6,088,628 6,097,169 6,114,824 6,124,693 6,140,788 6,141,863 6,157,155 6,160,324 6,163,124 6,177,650 6,180,898 6,181,096 6,188,194 6,208,105 6,212,444 6,219,583 6,226,181 6,236,011 6,236,896 6,250,174 6,278,902 6,279,413 6,285,921 6,298,283 6,321,139 6,324,443 6,328,523 6,330,493 6,340,875 6,356,671 6,377,869 6,382,012 6,384,371 6,396,030 6,414,711 6,424,883 6,431,018 6,434,448 6,445,979 6,459,958 6,463,358 6,484,067 6,486,629 6,507,165 6,654,666 6,665,588 6,680,461 6,696,810 6,728,417 6,763,284 6,772,493 6,845,296 6,853,881 6,888,089 6,898,486 6,917,837 6,928,337 6,965,091 6,970,802 7,038,165 7,069,808 7,084,900 7,092,791 7,133,747 7,143,100 7,149,602 7,131,848 7,161,321 7,171,041 7,174,234 7,173,213 7,177,722 7,177,439 7,181,294 7,181,313 7,280,687 7,283,661 7,291,806 7,299,713 7,315,650 7,324,873 7,328,083 7,330,777 7,333,879 7,355,725 7,359,817 7,373,220 7,376,488 7,386,367 7,464,623 7,447,615 7,445,260 7,474,939 7,486,816 7,495,192 7,501,778 7,502,504 7,508,155 7,512,459 7,525,273 7,526,121

Conventions

WARNING
Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION
Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system–including the design, implementation, and debugging of application programs–must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti–tie–down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end–use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro–magnetic interference (EMI), radio frequency interference (RFI), and electro–static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

  - Using a low motion speed, single step the program for at least one full cycle.
  - Using a low motion speed, test run the program continuously for at least one full cycle.
  - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3–phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back–check limit switches in the workcell to make sure they do not fail.
- Implement ‘‘failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high–voltage electrostatic tools.

CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

WARNING

Observe all safety rules and guidelines to avoid injury.

WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom–shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.

1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
4. When the dust has been removed, the covers must be replaced immediately.
5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding**: All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage**: High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the ‘‘home’’ position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
  - Wear eye protection and protective gloves during application and removal.
  - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
  - If there is contact with the skin, wash with water.
  - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
  - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
  - Adequate ventilation is required. Overexposure could cause drowsiness.
  - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
  - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

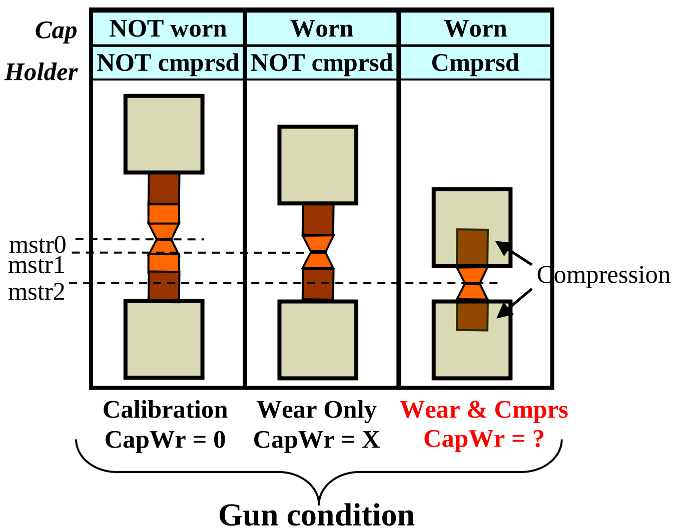
TABLE OF CONTENTS

1 OVERVIEW

1. OVERVIEW

1.1 Background

Standard Tipwear compensation function (included with servogun option) only tracks TCPDiff, cap wear and compression are not tracked.
Cap Wear value does not equal TcpDiff, if displacement changes (i.e. non-cap wear changes like compression) are present, as shown below.



Limitation of standard tip wear compensation function (without R826)

TcpDiff = wear.fix + comprs.fix

If compression = 0
CapWear.fix = TcpDiff
CapWear.mov = TotalWear – TcpDiff

If compression <> 0
CapWear.fix = ?
CapWear.mov = ?

Tip wear diagnostics option (R826) eliminates this limitation and provides the following functionality:

- tracking of cap wear, compression, and tip installation
- historical db/archive of wear measurements
- tools and visualizations to analyze guntip changes

See appendix for detailed function list

Motivation:

- cap wear value must be measured to decide whether cap replacement is needed
- unusual cap wear rates are indication of weld problems
- compression changes are indication of gun problem (for example belt slippage or shank damage)

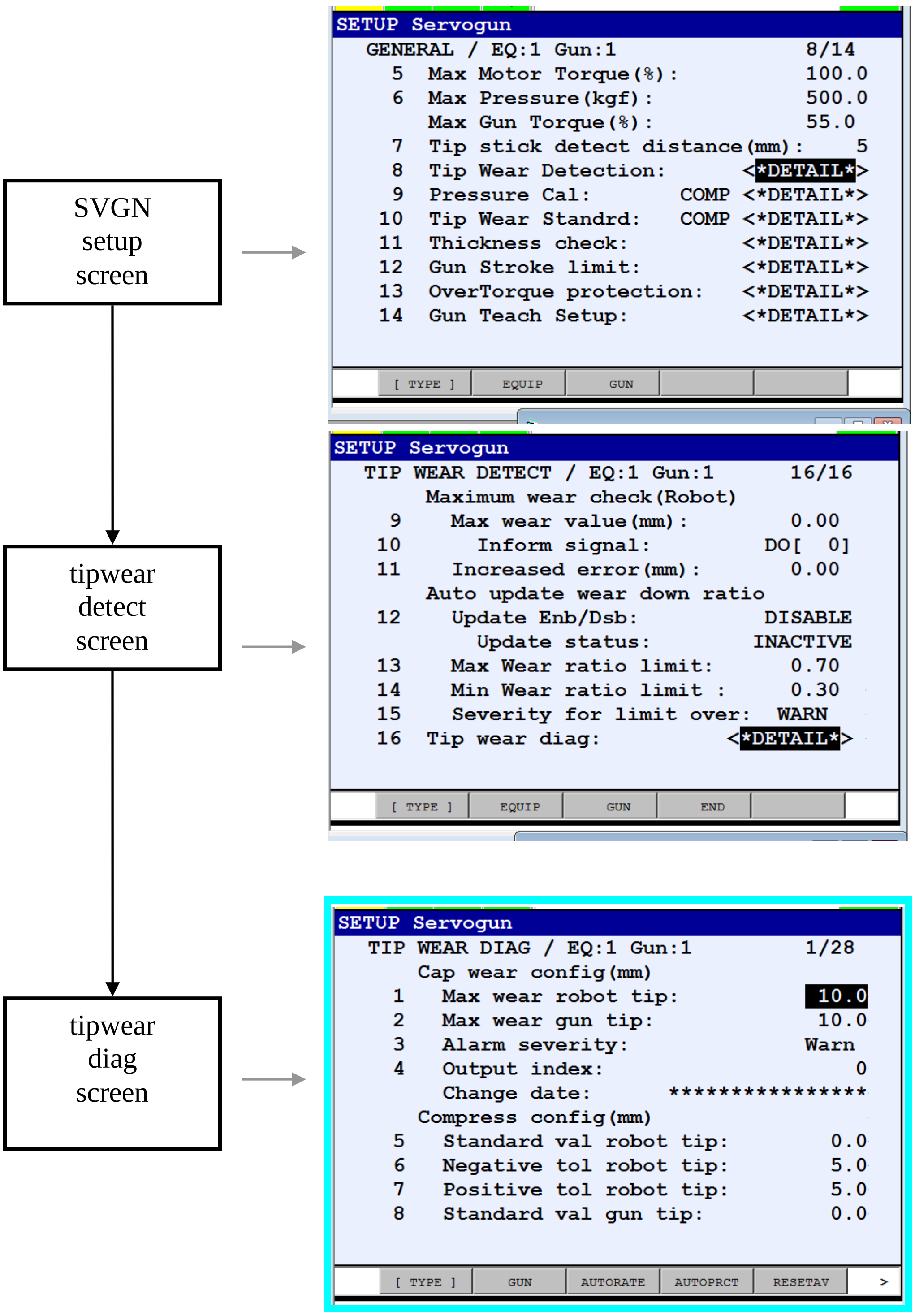
Terms	Description
Wear	Cap erosion or material consumed during weld process
Compression	Displacement of cap into shank or anything that moves tip surface in cap wear direction
TCP diff	Difference between tip surface (Z location) at 1. tipwear calibration 2. time of measurement (tw_update)

2 TIP WEAR DIAG CONFIG

2. TIP WEAR DIAG CONFIG

This screen configures alarm and monitoring parameters for tip wear. Access this screen as shown below.

2.1 Screen selection roadmap



2.2 Parameters

#	Item	Description
	CapWear config	
1	Max wear robot tip	If robot tip capwear exceeds max wear robot tip, SVGN-423 Fixed tip capwear: #.# > limit
2	Max wear gun tip	If gun tip capwear exceeds max wear gun tip, SVGN-424 Movable tip capwear: #.# > limit
3	Alarm severity	Severity of SVGN-423, SVGN-424 alarm: none, warn or pause
4	Output index	DOUT# asserted at capwear alarm
	Change date	Date of most recent change to cap wear config parameter(s)
	Compress config(mm)	
5	Standard val robot tip	Expected distance that cap is pushed into holder AFTER CALIBRATION. Default=0. Change this value to accommodate a different cap holder, for example.
6	Negative tol robot tip	Max difference from standard value, in negative direction. IF compression < (standard value – negative tol) SVGN-429 Fixed tip compression: #.# overtol
7	Positive tol robot tip	Max difference from standard value, in positive direction. IF compression > (standard value + positive tol) SVGN-429 Fixed tip compression: #.# overtol
8	Standard val gun tip	Expected distance that cap is pushed into holder AFTER CALIBRATION. Default =0. Change this value to accommodate a different cap holder, for example.
9	Negative tol gun tip	Max difference from standard value, in negative direction. IF compression < (standard value – negative tol) SVGN-430 Movable tip compression: #.# overtol
10	Positive tol gun tip	Max difference from standard value, in positive direction. IF compression > (standard value + positive tol) SVGN-430 Movable tip compression: #.# overtol
11	Alarm severity	Severity of SVGN-429, SVGN-430 alarm: none, warn or pause
12	Output index	DOUT# asserted at compression overtol alarm
	Change date	Date of most recent change to compress config parameter(s)
	Wear rate config(1000*mm/spot)	
13	Standard val robot tip	Expected wear rate for robot tip.
14	Negative tol robot tip	Max difference from standard wear rate in negative direction. IF compression < (standard value – negative tol) SVGN-425 fixed tip wear rate: #.# overtol
15	Positive tol robot tip	Max difference from standard wear rate in positive direction. IF compression > (standard value + positive tol) SVGN-425 fixed tip wear rate: #.# overtol
16	Standard val gun tip	Expected rate of wear for gun tip.
17	Negative tol gun tip	Max difference from standard wear rate in negative direction. IF compression < (standard value – negative tol) SVGN-426 Movable tip wear rate: #.# overtol
18	Positive tol gun tip	Max difference from standard wear rate in positive direction. IF compression > (standard value + positive tol) SVGN-426 Movable tip wear rate: #.# overtol
19	Alarm severity	Severity of SVGN-425, SVGN-426 alarm: none, warn or pause

20	Output index	DOUT# asserted at wear rate overtol alarm
	Change date	Date of most recent change to wear rate config parameter(s)
	Wear percent config (1000*mm/spot)	
21	Standard val robot tip	Expected wear percentage. Default=50.
22	Negative tol robot tip	Max difference from standard wear percentage, in negative direction. IF wear percent < (standard value – negative tol) SVGN-427 Fixed tip wear % #.# overtol
23	Positive tol robot tip	Max difference from standard wear percentage, in positive direction. IF wear percent > (standard value + positive tol) SVGN-427 Fixed tip wear % #.# overtol
24	Standard val gun tip	Expected wear percentage. Default=50.
25	Negative tol gun tip	Max difference from standard wear percentage, in negative direction. IF wear percent < (standard value – negative tol) SVGN-428 Movable tip wear % #.# overtol (x = wear percentage)
26	Positive tol gun tip	Max difference from standard wear percentage, in positive direction. IF wear percent > (standard value + positive tol) SVGN-428 Movable tip wear % #.# overtol (x = wear percentage)
27	Alarm severity	Severity of SVGN-427,428 alarm: none, warn or pause
28	Output index	DOUT# asserted at wear percent overtol alarm
	Change date	Date of most recent change to wear% config parameter(s)

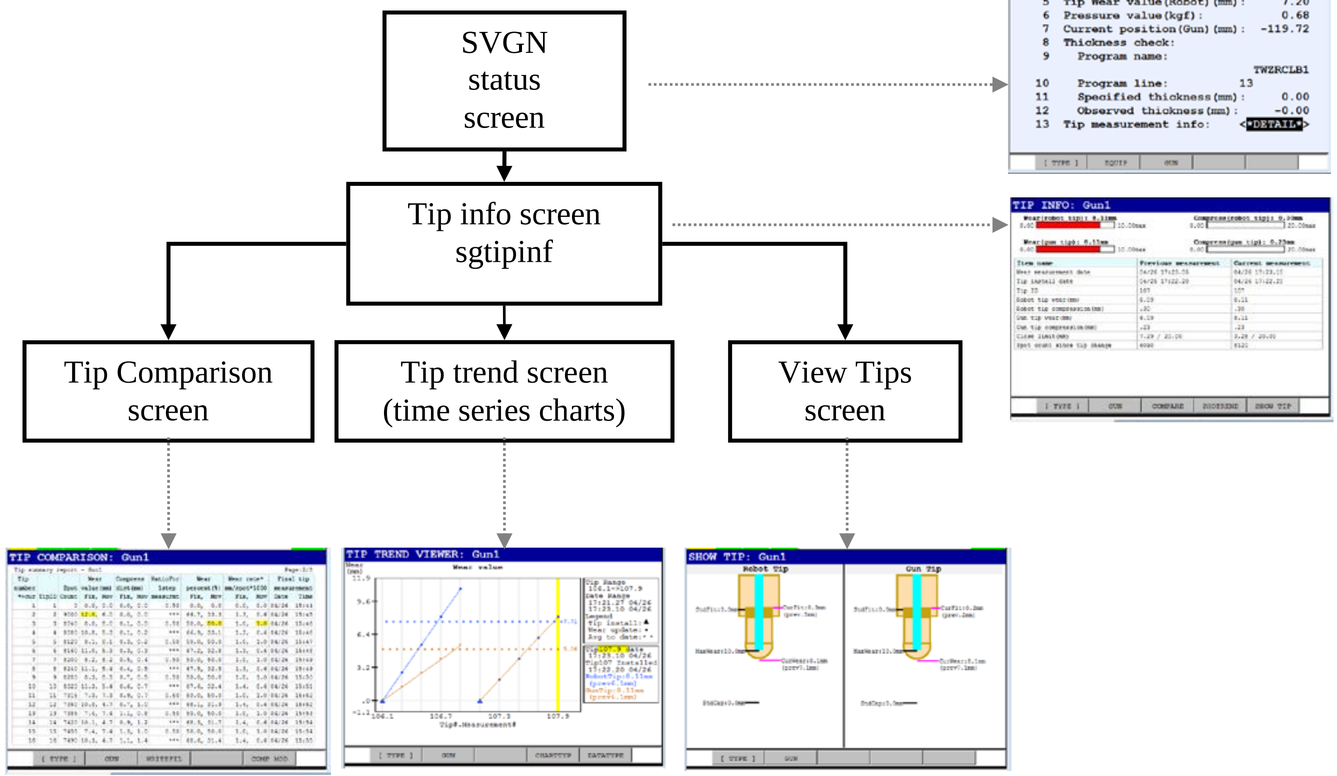
2.3 Fkeys

Fkey	Caption	Operation
F3	AutoRate	Automatically set wear rate std and tolerance parameters (based on current averages) $twgun#.$wearratcfg.$std_val_fix = $twgun#.$wearratavg.$fixedtip $twgun#.$wearratcfg.$fix_tol_pos = $twsyscfg.$cfgtol_neg * $twgun#.$wearratcfg.$std_val_fix $twgun#.$wearratcfg.$fix_tol_neg = $twsyscfg.$cfgtol_pos * $twgun#.$wearratcfg.$std_val_fix $twgun#.$wearratcfg.$std_val_mov = $twgun#.$wearratavg.$movabletip $twgun#.$wearratcfg.$mov_tol_pos = $twsyscfg.$cfgtol_neg * $twgun#.$wearratcfg.$std_val_mov $twgun#.$wearratcfg.$mov_tol_neg = $twsyscfg.$cfgtol_pos * $twgun#.$wearratcfg.$std_val_mov Note: if num_terms in average = 0, the following message is posted: “Wear rate parms not set, average = 0 “
F4	AutoPrct	Automatically set wear rate percent and tolerance parameters (based on current averages) $twgun#.$wearpctcfg.$std_val_fix = $twgun#.$wearpctavg.$fixedtip $twgun#.$wearpctcfg.$fix_tol_pos = $twsyscfg.$cfgtol_neg * $twgun#.$wearpctcfg.$std_val_fix $twgun#.$wearpctcfg.$fix_tol_neg = $twsyscfg.$cfgtol_pos * $twgun#.$wearpctcfg.$std_val_fix $twgun#.$wearpctcfg.$std_val_mov = $twgun#.$wearpctavg.$movabletip $twgun#.$wearpctcfg.$mov_tol_pos = $twsyscfg.$cfgtol_neg * $twgun#.$wearpctcfg.$std_val_mov $twgun#.$wearpctcfg.$mov_tol_neg = $twsyscfg.$cfgtol_pos * $twgun#.$wearpctcfg.$std_val_mov Note: if num_terms in average = 0, the following message is posted: “Wear percent parms not set, average = 0 “

F5	ResetAvg	Reset average to date. $twgun#.$wearendavg1 = 0, $twgun#.$wearendavg2 = 0 $twgun#.$wearratavg1 = 0, $twgun#.$wearratavg2 = 0 $twgun#.$wearpctavg1 = 0, $twgun#.$wearpctavg2 = 0 $twgun#.$cmprsavg1 = 0, $twgun#.$cmprsavg2 = 0 $twgun#.$cmprsratavg1 = 0, $twgun#.$cmprsratavg2 = 0 Note (parm1 = fixed tip, parm2 = movable tip)
F6	ResetMax	Reset max value to date for all parameters. $twgun#.$wearendmax1 = 0, $twgun#.$wearendmax2 = 0 $twgun#.$wearratmax1 = 0, $twgun#.$wearratmax2 = 0 $twgun#.$wearpctmax1 = 0, $twgun#.$wearpctmax2 = 0 $twgun#.$cmprsmax1 = 0, $twgun#.$cmprsmax2 = 0 $twgun#.$cmprsratmax1 = 0, $twgun#.$cmprsratmax2 = 0 Note (parm1 = fixed tip, parm2 = movable tip)

3 TIP WEAR DIAG STATUS

3. TIP WEAR DIAG STATUS



<u>All screens display raw wear and compression measurement data</u>

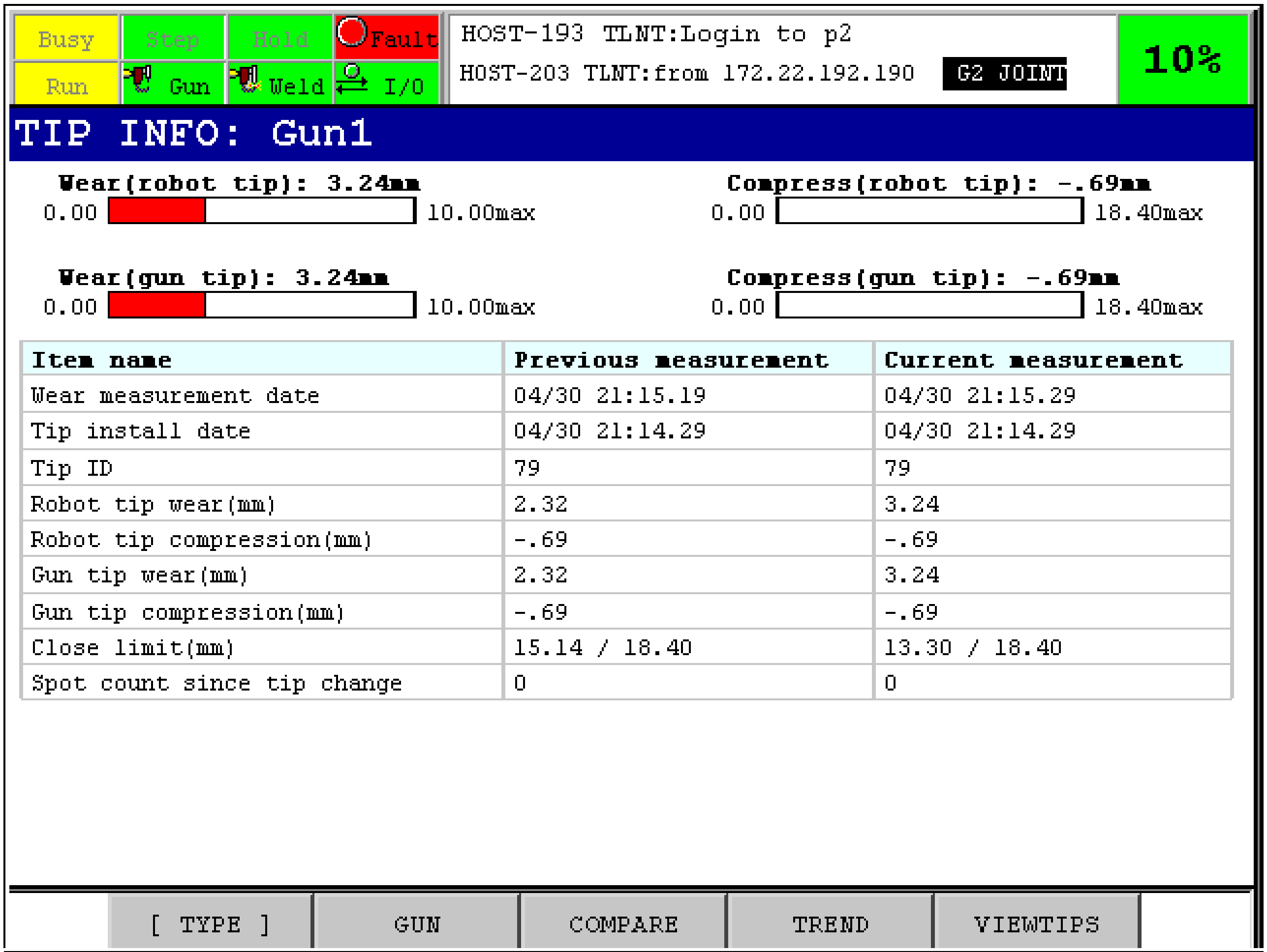
- fixed tip wear
- movable tip wear
- fixed tip compression
- movable tip compression
- commanded wear ratio
- measurement type (1/2 step)
- spot count

<u>Comparison screen and trend screen display additional statistics (computed from raw data)</u>

- wear rate (wear mm / spot)
- wear ratio (percent of total wear)
- compression rate (mm/tip

3.1.1 Tip info screen (main)

This screen is used to access all tipwear diag functions (via fkeys).
Screen itself is used to review **current** tip wear status in tabular format.
Use VIEW TIP screen for graphical description of current status.
Use TIP TREND screen and TIP COMPARE screen for detailed review and diagnostics.



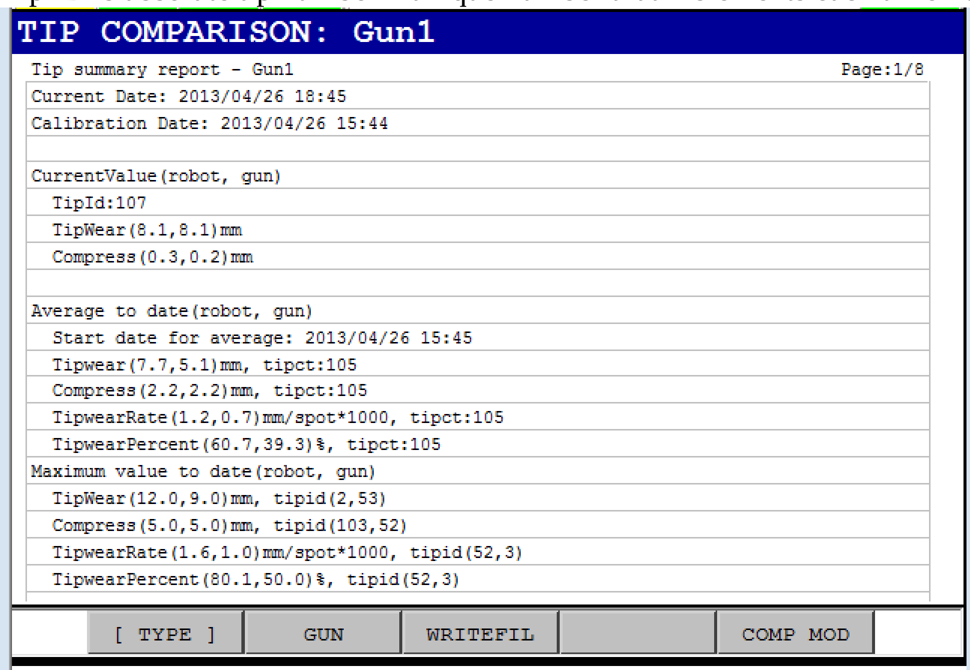
Note:
tipid: tip deployment number - incremental number assigned at new tip install. At tipwear cal, tipid = 1. First tip installed after calibration is tipid=2.

3.1.2 Compare screen

Use this screen to review tip data and compare tips. Several comparison modes are available.

3.1.2.1 Compare mode1: Tip summary

In this mode, cumulative values are presented on first page.
Final measurment value for each tip is presented on subsequent pages.
Note that tip number is number in archive (first tip number always equals 1)
Tip ID is absolute tip number – unique number that increments each time new tip is installed.



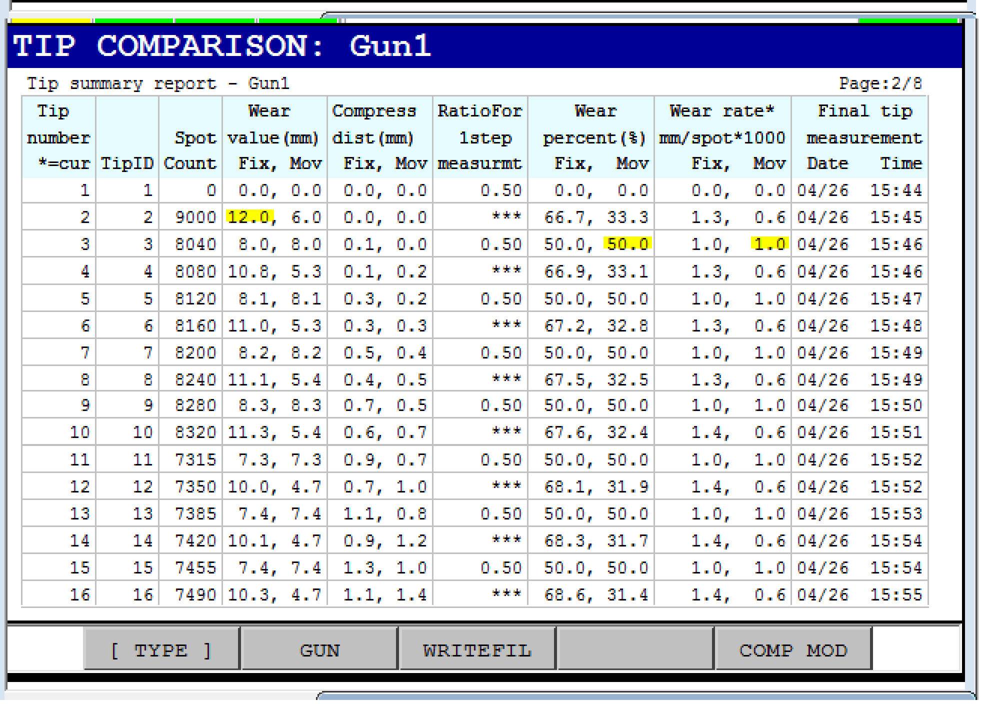
TIP COMPARISON: Gun1

Tip summary report - Gun1 Page:1/8
Current Date: 2013/04/26 18:45
Calibration Date: 2013/04/26 15:44

CurrentValue(robot, gun)
TipId:107
TipWear(8.1,8.1)mm
Compress(0.3,0.2)mm

Average to date(robot, gun)
Start date for average: 2013/04/26 15:45
Tipwear(7.7,5.1)mm, tipct:105
Compress(2.2,2.2)mm, tipct:105
TipwearRate(1.2,0.7)mm/spot*1000, tipct:105
TipwearPercent(60.7,39.3)%, tipct:105
Maximum value to date(robot, gun)
TipWear(12.0,9.0)mm, tipid(2,53)
Compress(5.0,5.0)mm, tipid(103,52)
TipwearRate(1.6,1.0)mm/spot*1000, tipid(52,3)
TipwearPercent(80.1,50.0)%, tipid(52,3)

[TYPE] GUN WRITEFIL COMP MOD



TIP COMPARISON: Gun1

Tip summary report - Gun1 Page:2/8

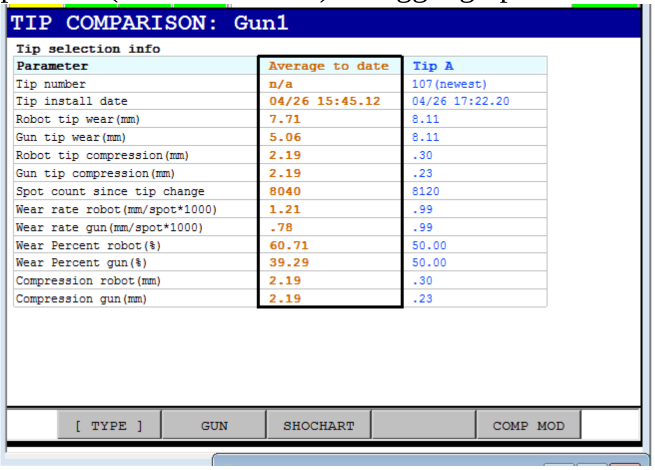
Tip number *=cur	TipID	Spot Count	Wear value(mm) Fix, Mov	Compress dist(mm) Fix, Mov	RatioFor 1step measurmt	Wear percent(%) Fix, Mov	Wear rate* mm/spot*1000 Fix, Mov	Final tip measurement Date Time
1	1	0	0.0, 0.0	0.0, 0.0	0.50	0.0, 0.0	0.0, 0.0	04/26 15:44
2	2	9000	12.0, 6.0	0.0, 0.0	***	66.7, 33.3	1.3, 0.6	04/26 15:45
3	3	8040	8.0, 8.0	0.1, 0.0	0.50	50.0, 50.0	1.0, 1.0	04/26 15:46
4	4	8080	10.8, 5.3	0.1, 0.2	***	66.9, 33.1	1.3, 0.6	04/26 15:46
5	5	8120	8.1, 8.1	0.3, 0.2	0.50	50.0, 50.0	1.0, 1.0	04/26 15:47
6	6	8160	11.0, 5.3	0.3, 0.3	***	67.2, 32.8	1.3, 0.6	04/26 15:48
7	7	8200	8.2, 8.2	0.5, 0.4	0.50	50.0, 50.0	1.0, 1.0	04/26 15:49
8	8	8240	11.1, 5.4	0.4, 0.5	***	67.5, 32.5	1.3, 0.6	04/26 15:49
9	9	8280	8.3, 8.3	0.7, 0.5	0.50	50.0, 50.0	1.0, 1.0	04/26 15:50
10	10	8320	11.3, 5.4	0.6, 0.7	***	67.6, 32.4	1.4, 0.6	04/26 15:51
11	11	7315	7.3, 7.3	0.9, 0.7	0.50	50.0, 50.0	1.0, 1.0	04/26 15:52
12	12	7350	10.0, 4.7	0.7, 1.0	***	68.1, 31.9	1.4, 0.6	04/26 15:52
13	13	7385	7.4, 7.4	1.1, 0.8	0.50	50.0, 50.0	1.0, 1.0	04/26 15:53
14	14	7420	10.1, 4.7	0.9, 1.2	***	68.3, 31.7	1.4, 0.6	04/26 15:54
15	15	7455	7.4, 7.4	1.3, 1.0	0.50	50.0, 50.0	1.0, 1.0	04/26 15:54
16	16	7490	10.3, 4.7	1.1, 1.4	***	68.6, 31.4	1.4, 0.6	04/26 15:55

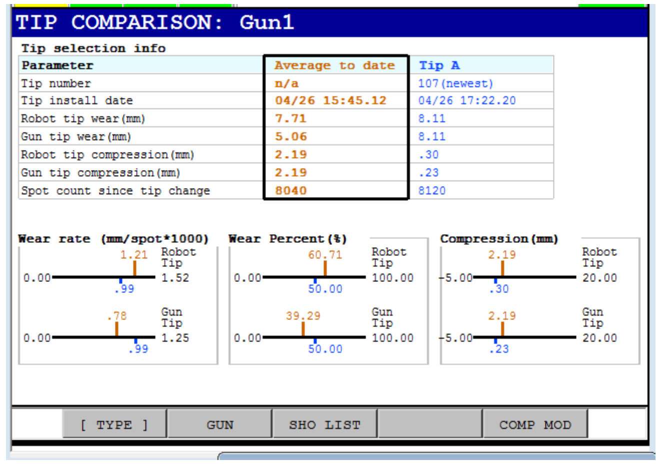
[TYPE] GUN WRITEFIL COMP MOD

3.1.2.2 Compare mode2: Average vs. Tip A

This screen compares values for selected tip vs average.
Average to date or archive average can be displayed – select left column, and press up/down arrow to toggle between archive average and average to date. Select right column and press up/down arrow to change tip selection.

press F3(shochart/sholist) to toggle graphic charts display(3 line charts at bottom of screen)

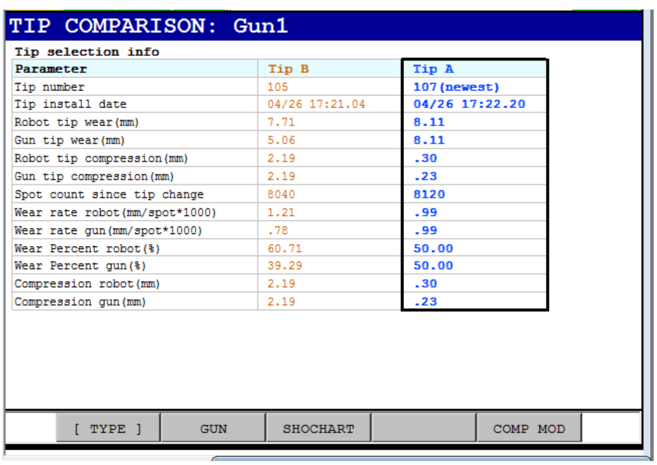


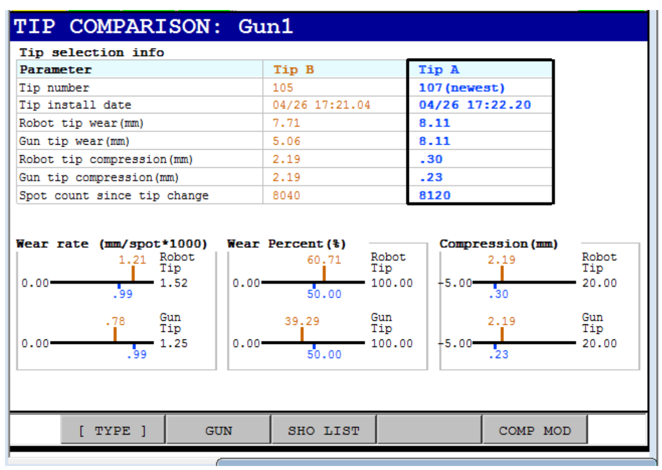


3.1.2.3 Compare mode3: TipB vs. Tip A

This screen compares values for selected tip vs average.
select left column, and press up/down arrow to change tip B selection
Select right column and press up/down arrow to change tip A selection.

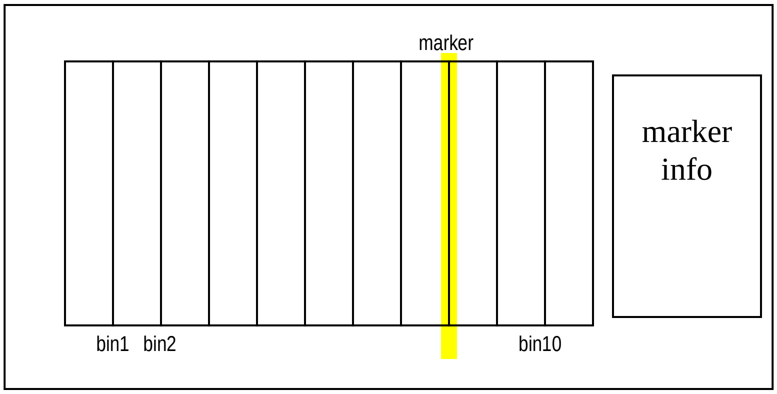
press F3(shochart/sholist) to toggle graphic charts display(3 line charts at bottom of screen)





3.1.3 Tip Trend screen
This screen displays trend data (series of measurements) for wear and other parameters.

3.1.3.1 Trend screen overview



Notes:
- chart displays 10 bins
- bin content varies with parmtyp and charttype
- std bin = 1 wear measurement
- trend bin = N end measurements
- meas1 = new tip (1st measurement)
- measurement N = end (final measurement)
- Line showing average to date displays on each chart (toggle this line by pressing enter key)

The following statistical parameters are available on trend screen:
. Absolute wear value (mm)
. Wear Rate (tip erosion per spot)
. Wear Ratio observed
. Compression value (mm)
. Compression rate (mm/tip)

Data for statistical parameters is available in 3 formats:

Format	**data type**	**points per bin**	**Comments**
standard	wear	1	Each chart point (bin) corresponds to tipwear measured at wear update.
standard	other	1	Each chart point (bin) corresponds to specified parameter measured at wear update end (final wear measurement before tip change).
condensed, averaged	all	endmeas ct / 10	Each chart point (bin) corresponds to a series of the specified parameter measured at wear update end measurement. This point is the average for a group of measurements. For example if archive contains 100 wear update end measurements, then: bin1 = average of meas1, meas2,…, meas10 bin2 = average of meas10, meas11,…, meas20 : bin10 = average of meas90, meas91,…, meas100
condensed, scattered	all	endmeas.ct / 10	Each chart point (bin) corresponds to the specified parameter measured at wear update end measurement. Multiple points are plotted at each bin. For example if archive contains 100 wear update end measurements, then: bin1 = meas1, meas2,…, meas10 bin2 = meas10, meas11,…, meas20 : bin10 = meas90, meas91,…, meas100

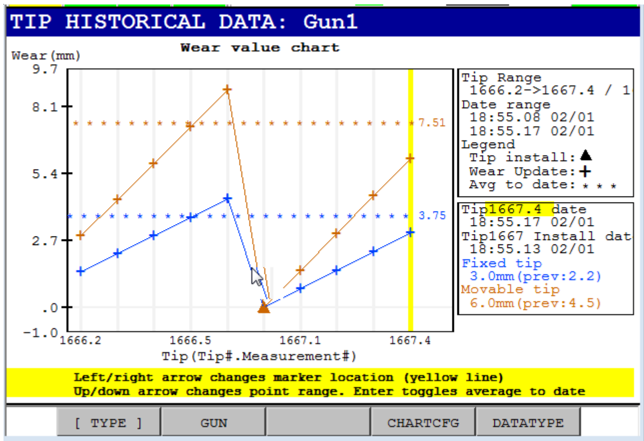
Note: “scatter” format is omitted in following pages.

3.1.3.2 Trend screen fkeys

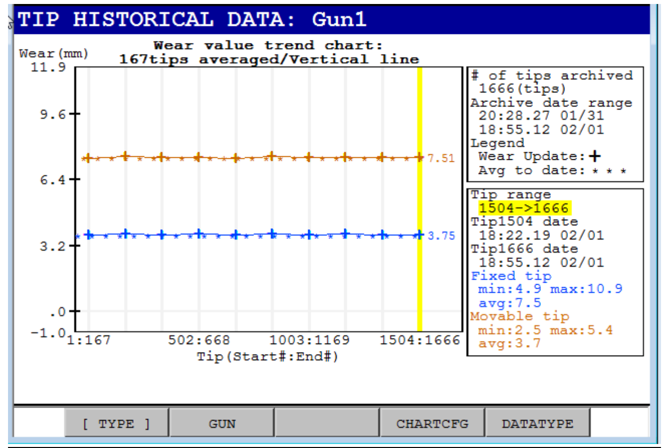
Fkey	Caption	Operation
F2	Gun	Pressing F2 displays dropdown list with guns available. Select desired gun#
F4	Chart Type	Pressing F4 displays dropdown list: 1: Standard Chart 2: Trend Chart Select desired chart type, per description on previous page
F5	Data Type	Pressing F5 displays dropdown list: 1: Wear value (mm) 2: Wear Rate (mm/spot) 3: Compression value (mm) 4: Compression value (mm) 5: Compress Rate (0.1mm/tip) Select desired chart type, per description on previous page

3.1.3.3 Chart1: Absolute tip wear value (mm)

Standard Chart

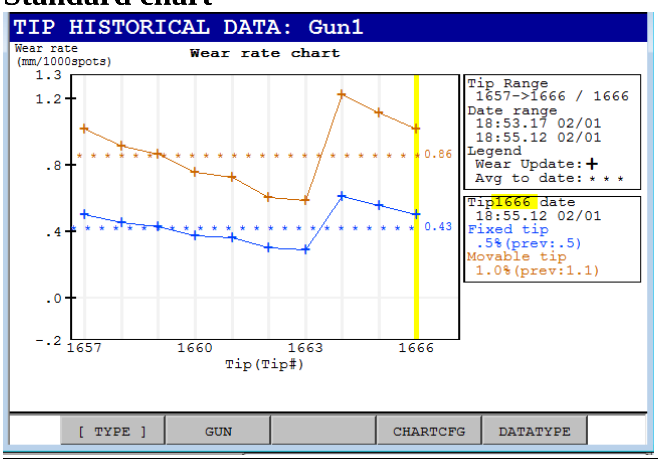


Trend Chart

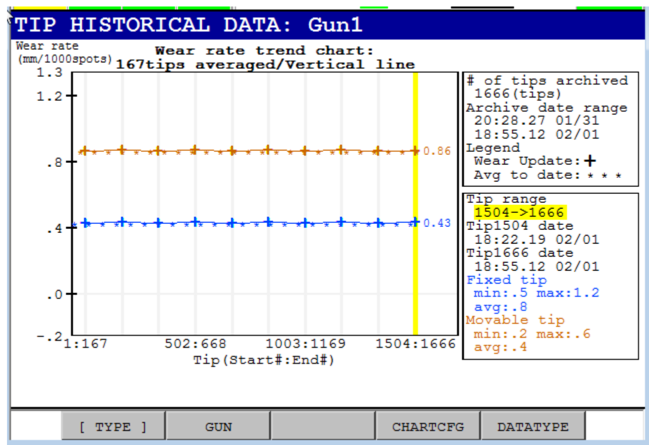


3.1.3.4 Chart2: Wear rate(mm)

Standard chart

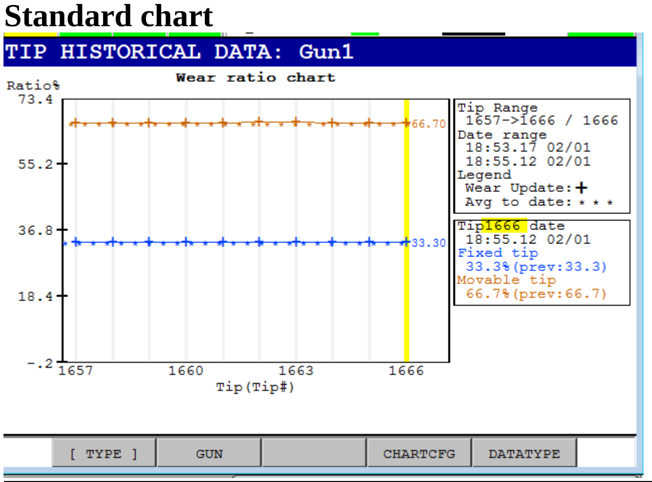


Trend Chart

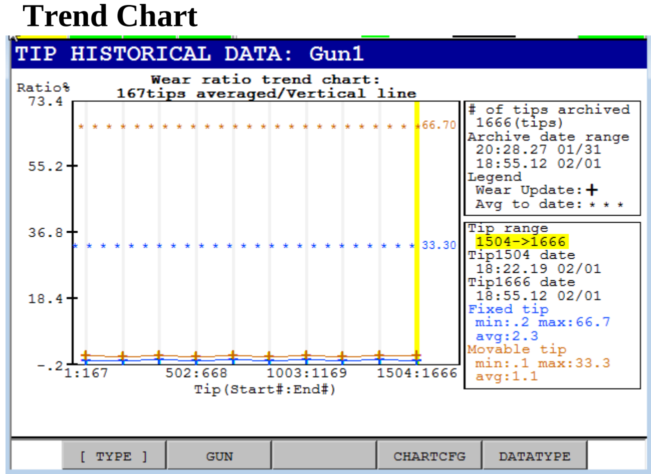


3.1.3.5 Chart3: Wear ratio observed

Standard chart

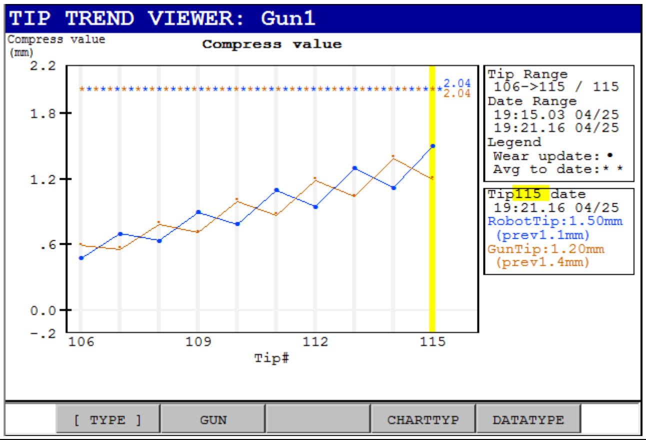


Trend Chart

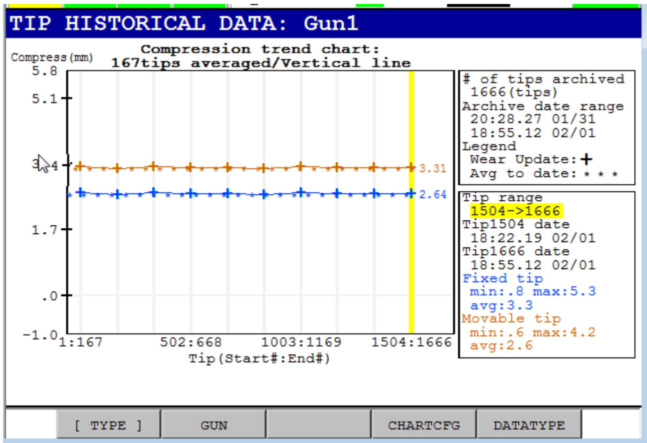


3.1.3.6 Chart4: Compression

Standard chart



Trend chart

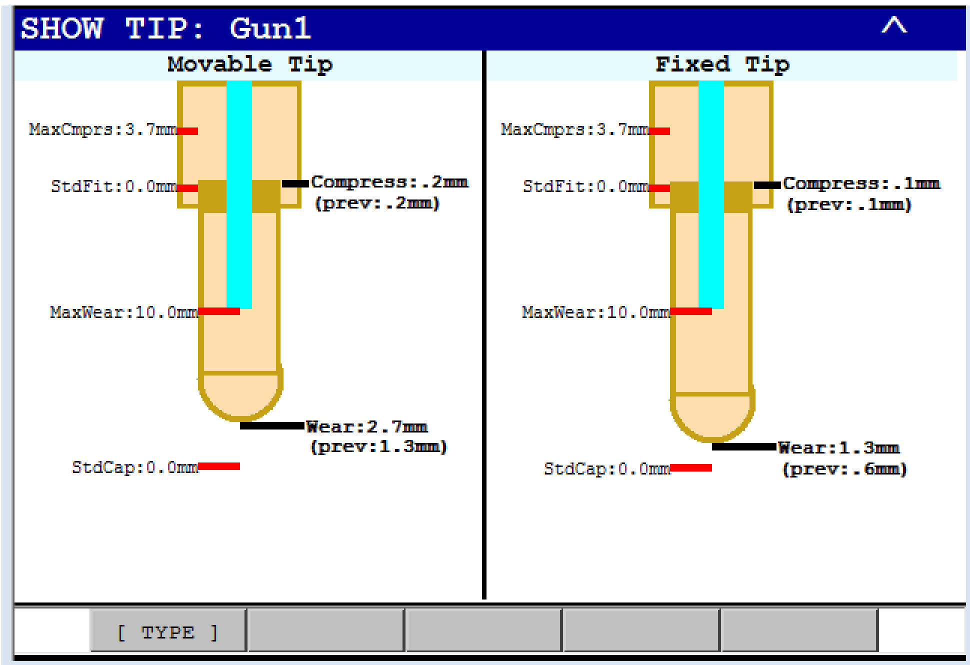


Note: compression rate is obtained from backward difference of points in archive. Therefore, tip1 always has compress rate = 0 (initial compression is 0, by definition).

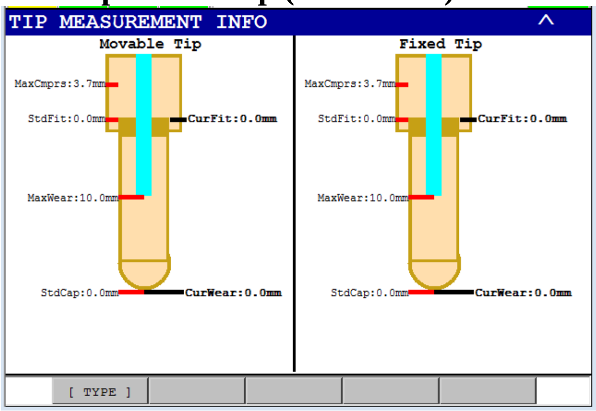
3.1.4 Show Tip screen

Use this screen to see a visual representation of current tip condition.
Cap wear or erosion is merely loss of cap material over time. Typical cap can erode approximately 7mm before water jacket (shown in blue) is reached. Alarm symbol is displayed, if wear exceeds this value.

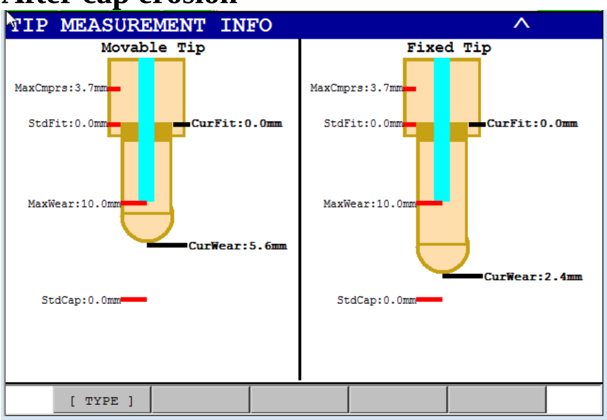
Typical gun electrode end has taper-fit opening. Over time, this opening becomes wider, so electrode insertion distance increases. Normal / non-enlarged position is indicated by standard fit. Compression is additional displacement of electrode holder.



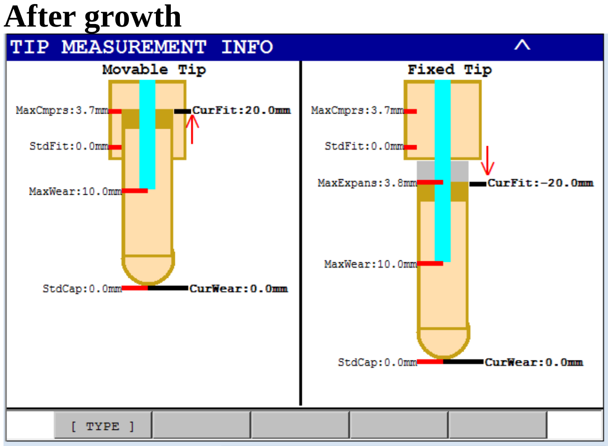
After tip wear setup (calibration)



After cap erosion



After growth



3.2 File backup

when file backup is performed, the following files are automatically added to gun#.zip.

archivls.txt

This is complete list of raw data in archive.

File is written with oldest archive record on top line, and newest archive record on bottom line.

```
   rec#,aridx, tipid,measct, wrfix, wrmov,cmprsfix,cmprsmov,meastyp,ratio1step,spotct, date, time
      1,    1,     1,     1,   0.0,   0.0,     0.0,     0.0,      1,      0.50,     0,07/09,16:35
      2,    2,     1,    16,   2.2,   2.2,     0.0,     0.0,      1,      0.50,     0,07/09,16:44
      3,    3,     1,    22,   4.0,   4.0,     0.0,     0.0,      1,      0.50,     0,07/09,16:45
      4,    4,     2,     1,   0.0,   0.0,     0.4,     0.4,      1,      0.50,     0,07/09,16:45
      5,    5,     2,     4,   0.8,   0.8,     0.4,     0.4,      1,      0.50,     0,07/09,16:45
```

tipsumry.txt

this is a text file with same text as tipsummary screen .

```
TIPSUMRY.TXT
Current Date: 2013/07/09 16:45
Calibration Date: 2013/07/09 16:35

CurrentValue(robot, gun)
   TipId:2
   TipWear(0.9,0.9)mm
   Compress(0.4,0.4)mm

Average to date(robot, gun)
   Start date for average: 2013/07/09 16:45
   Tipwear(4.0,4.0)mm, tipct:1
   Compress(0.0,0.0)mm, tipct:1
   TipwearRate(0.0,0.0)mm/spot*1000, tipct:0
   TipwearPercent(50.0,50.0)%, tipct:1
Maximum value to date(robot, gun)
   TipWear(4.0,4.0)mm, tipid(1,1)
   Compress(0.0,0.0)mm, tipid(0,0)
   TipwearRate(0.0,0.0)mm/spot*1000, tipid(0,0)
   TipwearPercent(50.0,50.0)%, tipid(1,1)

  Tip                 Wear    Compress  RatioFor     Wear     Wear rate*    Final tip
 number         Spot value(mm) dist(mm)    1step   percent(%) mm/spot*1000  measurement
  *=cur TipID Count  Fix, Mov  Fix, Mov measurmt   Fix,  Mov    Fix,  Mov  Date   Time
      1     1     0  4.0, 4.0  0.0, 0.0     0.50  50.0, 50.0    0.0,  0.0 07/09  16:45
     *2     2     0  0.8, 0.8  0.4, 0.4     0.50  50.0, 50.0    0.0,  0.0 07/09  16:45
```

4 APPENDIX

4. APPENDIX

4.1 Tip wear diagnostics feature summary

- track TCPDiff, CapWear AND compression(non-cap wear).
- provide notification – alarms and io – for abnormal conditions
- maintain historical database for Tip wear measurements and other data
- “smart save” feature to minimize storage (memory) requirements –record written only if one of the following conditions is met:
    - wear measurement is 1st for tip (newtip flag is set)
    - wear measurement is last for tip (newtip flag is set)
    - current wear measurement or compression > 1mm from record previously stored
- statistics for diagnosis and debug
    - wear value (mm): fixed tip, movable tip
    - wear rate (mm/spot) = erosion per spot
    - wear percent of total wear (ratio observed) =
        - fixed tip wear / total wear
        - movable tip wear / total wear
    - ratio setpoint, for 1step wear measurement
    - compression value (mm): fixed tip, movable tip
    - compression rate (mm/tip): fixed tip, movable tip
- track long-term (permanent average)
- data displays in multiple formats to reveal trends, problems, etc.
    - graphical status bar for quick review
    - trend line chart to see change over time
    - tip a vs. tip b comparison (side by side list)
    - complete line chart for complete review
    - “tip show” screen (caricature of tip state)

4.2 Limitations of Tip wear diagnostic option

Graphics and other tip wear diagnostic functions can be used with one or two step measurement. However, if one-step wear measurement is used, compression distribution is unknown, so equal compression is assumed. Two-step wear compensation must be used to identify compression distribution.

Compression change is only observable at newtip install. I.e. compression that occurs between tip install and tip replacement is indistinguishable from cap wear.

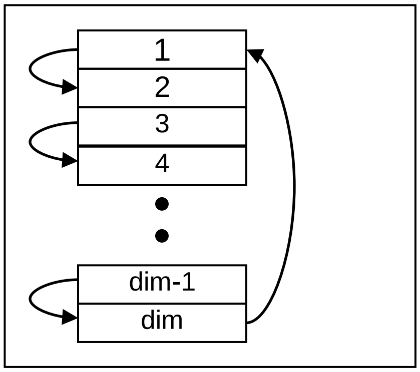
TWK macros (i.e. basic Fanuc tipwear function routines) must be used. Trend data, etc, is not available, if TWK macros are not used.

4.3 Archive function

Archive is simple queue.
Archive dimension specifies how may records are kept.

Memory is automatically created for archive at power-up. Archive records are updated when tip wear macros executes*. User does not have to perform any additional steps.

When archive is full, new record overwrites oldest record.



Default archive dimension is 500 records.
This means that 500 of measurements are available at any time. Archive size is specified via:
$TWSYSCFG.$ARCHIVDIMRQST (default = 500)

4.4 Archive update control

If tipwear is updated tipwear every job, saving measurement record in archives can lead to a small number of tips in the archive. This can be overcome by increasing archive dimension. However, larger archive increases memory usage. Smart save feature can be used to save significant wear measurements. i.e. measurements that differ more than say 2.0 mm from previous archive record.

Archive update can be disabled
by setting $TWSYSCFG.$ARCHIV_ENBL=FALSE (default =TRUE).

Archive update frequency is specified via:.
$TWSYSCFG.$ARCHIVSAVDIF (default = 2.0)
This means that tipwear update info (record) is archived if current wear measurement differs from previously saved wear measurement by >= 2.0 mm.
Note: first and last measurement for a tip are stored, regardless of value in $ARCHIVSAVDIF

4.5 Long-term tracking

Measurement archive provides tracking for each measurement performed. However, “average to date” values are very useful too. Average to date can provide a guideline for expected value of a given parameter. Similarly, maximum value to date can provide an indication of when parameter is abnormal.

Average to date is tracked for the following parameters:

wear end: $twgun#.$wearendavg1
wear rate: $twgun#.$wearratavg1
wear percent: $twgun#.$wearpctavg1
compression: $twgun#.$cmprsavg1
compression rate: $twgun#.$cmprsratavg1

Maximum to date is tracked for the following parameters:

wear end: $twgun#.$wearendavg1
wear rate: $twgun#.$wearratavg1
wear percent: $twgun#.$wearpctavg1
compression: $twgun#.$cmprsavg1
compression rate: $twgun#.$cmprsratavg1

4.6 Alarm db

SVGN	alarm string	cause	remedy
423	Fixed tip capwear:%s > limit.	Cap wear value exceeds max wear value.	Replace cap or increase $TWGUN.$WEARENDMAX.$FIXED_TIP
424	Movable tip capwear:%s > limit.	Cap wear value exceeds max wear value.	Replace cap or increase $TWGUN.$WEARENDMAX.$MOVABLE_TIP
425	Fixed tip wear rate:%s overtol.	Rate of wear exceeds tolerance.	Confirm that wear value is not excessive, for this number of spots or increase $TWGUN.$WEARRATMAX.$FIXED_TIP
426	Movable tip wear rate:%s overtol.	Rate of wear exceeds tolerance.	Confirm that wear value is not excessive, for this number of spots or increase $TWGUN.$WEARRATMAX.$MOVABLE_TIP
427	Fixed tip wear%%:%s overtol.	Wear percentage exceeds tolerance.	Confirm that spot process is operating correctly. or increase $TWGUN.$WEARPCTMAX.$FIXED_TIP
428	Movable tip wear%%:%s overtol.	Wear percentage exceeds tolerance.	Confirm that spot process is operating correctly or increase $TWGUN.$WEARPCTMAX.$MOVABLE_TIP
429	Fixed tip compression:%s overtol.	Gun tip compression (non-capwear change) exceeds tolerance.	Replace electrode / tip holder or increase $TWGUN.$COMPRESSMAX.$FIXED_TIP
430	Movable tip compression:%s overtol.	Gun tip compression (non-capwear change) exceeds tolerance.	Replace electrode / tip holder or increase $TWGUN.$COMPRESSMAX.$MOVABLE_TIP
431	Max wear not reachable(clslim=%s)	Close stroke limit is less than max wear value. Gun tip compression is too large and/or close limit is too small.	Increase stroke limit, if possible (if new limit does not exceed extension of leadscrew). Replace electrode / tip holder, if gun tip compression is large.
432	Nominal:%s, Tol(neg,pos):%s	Cause message posted with alarms 425-430 for additional info.	None - SVGN-432 is merely supplemental info for main alarm.
433	Insufficient memory for TWDIAG	$TWGUN#V#[] could not be created due to lack of memory.	Increase memory or reduce $TWSYSCFG.$ARCHIVDIMRQ.
434	Current wear (fix:%s,mov:%s)mm	Cause message posted with alarm 431 for additional info.	None - SVGN-434 is merely supplemental info for main alarm.

4.7 Sysvar list

$twsyscfg	data type	Description
$archivdimrqst	SHORT	size reqested for archive
$option_msk	SHORT	future use
$archivsavdif	float	min wear change for archive update0
$archiv_enbl	BOOLEAN	Enable writing to archive
$dram_floor	SHORT	Minimum memory floor for archive var creation
$infomsgmsk	float	Future
$cfgtol_neg	float	Negative tolerance for auto cfg val setting on alarm config screen
$cfgtol_pos	float	Positive tolerance for auto cfg val setting on alarm config screen

$twgun#	data type	Description
$ARCHIV_DIM	SHORT	dimension of archive actually created
$ARCHIV_CT	SHORT	current number of records in archive
$ARCHIV_IDX	SHORT	index of last record written to db (current measurement)
$IDX_4DIF	SHORT	internal var used for $archivsavdif function
$SPOTCT	SHORT	Number of spots fired since tip install (with weld enbl=TRUE)
$RESERVED2	SHORT	future use
$PLS_ZRO2	INTEGER	master count at new tip install
$OPN_STD2	REAL	gun open distance at new tip install
	TW_MEAS_T	
$MEAS.$MEASID	SHORT	Id of current measurement
$MEAS.$TIPID	SHORT	tip id of current tip
$MEAS.$MEASCT	SHORT	measurement count for current tip
$MEAS.$SPOTCT	SHORT	Cap wear of fixed tip
$MEAS.$CAPWEARFIX	REAL	Cap wear of movable tip
$MEAS.$CAPWEARMOV	REAL	Cap wear of fixed tip
$MEAS.$COMPRESSFIX	REAL	Cap wear of movable tip
$MEAS.$COMPRESSMOV	REAL	Compression of fixed tip
$MEAS.$WEARRATFIX	REAL	Compression of movable tip
$MEAS.$WEARRATMOV	REAL	Compression of fixed tip
$MEAS.$WEARPCTFIX	REAL	Wear percentage on fixed tip
$MEAS.$WEARPCTMOV	REAL	Wear percentage on movable tip
$MEAS.$CMPRSRATFIX	REAL	Compression rate, fixed tip
$MEAS.$CMPRSRATMOV	REAL	Compression rate, movable tip
$MEAS_MEM		copy of $meas, at wear update start
$WEARENDAVG.$FIXED_TIP	REAL	wear end value AVG to date, fixed tip
$WEARENDAVG.$MOVABLE_TIP	REAL	wear end value AVG to date, movable tip
$WEARENDAVG.$TIP_ID	SHORT	Tip id of tip that corresponds to wear end AVG to date
$WEARENDAVG.$RESERVED1	SHORT	future use
$WEARENDAVG.$TIMESTAMP	INTEGER	Timestamp of tip = wear end AVG to date
$WEARRATAVG.$FIXED_TIP	REAL	wear rate value AVG to date, fixed tip
$WEARRATAVG.$MOVABLE_TIP	REAL	wear rate value AVG to date, movable tip
$WEARRATAVG.$TIP_ID	SHORT	Tip id of tip that corresponds to AVG wear rate to date
$WEARRATAVG.$RESERVED1	SHORT	future use
$WEARRATAVG.$TIMESTAMP	INTEGER	Timestamp of tip = AVG wear end rate to date

$WEARPCTAVG.$FIXED_TIP	REAL	wear% value AVG to date, fixed tip
$WEARPCTAVG.$MOVABLE_TIP	REAL	wear% value AVG to date, movable tip
$WEARPCTAVG.$TIP_ID	SHORT	Tip id of tip that corresponds to AVG wear% to date
$WEARPCTAVG.$RESERVED1	SHORT	future use
$WEARPCTAVG.$TIMESTAMP	INTEGER	Timestamp of tip = to AVG wear% to date
$CMPRSAVG.$FIXED_TIP	REAL	compress value AVG to date, fixed tip
$CMPRSAVG.$MOVABLE_TIP	REAL	compress value AVG to date, movable tip
$CMPRSAVG.$TIP_ID	SHORT	Tip id of tip that corresponds to AVG compress to date
$CMPRSAVG.$RESERVED1	SHORT	future use
$CMPRSAVG.$TIMESTAMP	INTEGER	Timestamp of tip = to AVG compress to date
$CMPRSRATAVG.$FIXED_TIP	REAL	compress rate AVG to date, fixed tip
$CMPRSRATAVG.$MOVABLE_TIP	REAL	compress rate max to date, movable tip
$CMPRSRATMAX.$TIP_ID	SHORT	Tip id of tip that corresponds to max compress rate to date
$CMPRSRATMAX.$RESERVED1	SHORT	future use
$CMPRSRATMAX.$TIMESTAMP	INTEGER	Timestamp of tip = to max compress rate to date
$WEARENDMAX.$FIXED_TIP	REAL	wear end value max to date, fixed tip
$WEARENDMAX.$MOVABLE_TIP	REAL	wear end value max to date, movable tip
$WEARENDMAX.$TIP_ID	SHORT	Tip id of tip that corresponds to wear end max to date
$WEARENDMAX.$RESERVED1	SHORT	future use
$WEARENDMAX.$TIMESTAMP	INTEGER	Timestamp of tip = wear end max to date
$WEARRATMAX.$FIXED_TIP	REAL	wear rate value max to date, fixed tip
$WEARRATMAX.$MOVABLE_TIP	REAL	wear rate value max to date, movable tip
$WEARRATMAX.$TIP_ID	SHORT	Tip id of tip that corresponds to max wear rate to date
$WEARRATMAX.$RESERVED1	SHORT	future use
$WEARRATMAX.$TIMESTAMP	INTEGER	Timestamp of tip = max wear end rate to date
$WEARPCTMAX.$FIXED_TIP	REAL	wear% value max to date, fixed tip
$WEARPCTMAX.$MOVABLE_TIP	REAL	wear% value max to date, movable tip
$WEARPCTMAX.$TIP_ID	SHORT	Tip id of tip that corresponds to max wear% to date
$WEARPCTMAX.$RESERVED1	SHORT	future use
$WEARPCTMAX.$TIMESTAMP	INTEGER	Timestamp of tip = to max wear% to date
$CMPRSMAX.$FIXED_TIP	REAL	compress value max to date, fixed tip
$CMPRSMAX.$MOVABLE_TIP	REAL	compress value max to date, movable tip
$CMPRSMAX.$TIP_ID	SHORT	Tip id of tip that corresponds to max compress to date
$CMPRSMAX.$RESERVED1	SHORT	future use
$CMPRSMAX.$TIMESTAMP	INTEGER	Timestamp of tip = to max compress to date
$CMPRSRATMAX.$FIXED_TIP	REAL	compress rate max to date, fixed tip
$CMPRSRATMAX.$MOVABLE_TIP	REAL	compress rate max to date, movable tip
$CMPRSRATMAX.$TIP_ID	SHORT	Tip id of tip that corresponds to max compress rate to date
$CMPRSRATMAX.$RESERVED1	SHORT	future use
$CMPRSRATMAX.$TIMESTAMP	INTEGER	Timestamp of tip = to max compress rate to date
$CALIB_DATE	INTEGER	future use
$EXTRA_INT1	INTEGER	future use
$EXTRA_INT2	INTEGER	future use
$EXTRA_REAL1	REAL	future use
$EXTRA_REAL2	REAL	future use

$twgun#v database (each variable is array of short, with dim = $twsyscfg.$ardim_rqstd)

Name	Description
$twgun#v1	fixed tip wear * 100
$twgun#v2	movable tip wear * 100
$twgun#v3	fixed tip compression * 100
$twgun#v4	movable tip compression * 100
$twgun#v5	Measurement id
$twgun#v6	dostime ushort, lower
$twgun#v7	dostime ushort, upper
$twgun#v8	spot count
$twgun#v9	tip id (deployment number)
$twgun#v10	measurement count
$twgun#v11	measurement type
$twgun#v12	wear ratio command (1step)